Lehigh Valley Dynamometer Car Test.

Editor:

I have noticed several articles in your pages during the last few years concerning the building and operating of dynamometer test cars. As I have had considerable experience in that line in the last year I take the liberty of sending you an outline of the policy employed by the Lehigh Valley Railroad in an elaborate test recently made by them. In this test the aim was to get a reliable record of the draw-bar pull, together with the locations, speed, and steam pressure, rather than such minute details as the exact valve travel, the exact amount of water used, the weight of the ashes, and a thousand and one similar details, the exact knowledge of which has very little bearing on the moving of trains in the practical every-day runs. A division of a railroad system is usually subdivided into a number of routes for through freight trains.

There is always a spot on each of these routes where the trains draw harder than at any other point, on the route (in the same direction). If the engine can move the train over this point, then, the engine may usually be depended upon to move the train all the way without trouble. In making a test with a dynamometer car we may locate this point, on the record and divide the tractive effort required to move the train over the hardest pull, by the tonnage of that particular train, and thus obtain a constant, which will be the number of pounds required to pull one ton over the hardest pull at the same speed the train moved over that point in the trip.

This statement may be made more plain, thus:

$$\frac{\text{Tractive effort for train}}{\text{Tonnage of train}} = \text{lbs. per ton.}$$

The proper tonnage rating may then be reckoned, thus:

$$\frac{\text{Tractive effort of engine}}{\text{Pounds required for one ton}} = \text{tonnage}$$

for route and that engine.

We may also check up the tractive effort of the engine by comparing the measured tractive effort with the designed tractive effort, and thus determine the necessary reduction to be made from the mathematical or calculated tractive effort of an engine for this particular route. Of course an engine will not exert as great a draw-bar pull in rounding a curve, as it would be capable of doing on a straight track. Next to be kept in mind during the test, and the subsequent making out reports from same, is the amount of coal consumed by the different classes of engines per unit of power delivered.

The Lehigh Valley tests were made in five months, and covered their entire system. The test referred to above was in charge of the writer.

Sayre, Pa. W. S. Drummond.

Wheel Press Makes Good.

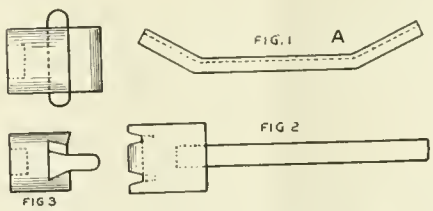
Editor:

Since the introduction of the all steel car there is quite a multitude of channel and I-beams and various other shapes used on these same cars that are exceedingly hard to bend and at the same time keep straight if they are heated. Even when heated and pressed by proper forms in a bulldozer they frequently get out of shape when cooling.

I found this out to my sorrow when I tried it once upon a time and had to rack my brain for some time to provide a method of bending, cold. The pieces were 12-in. channels. We had to keep a gang around the bulldozer with crow bars to skid the machine back on the foundation after having bent a few, cold, on it.

The bars had to be bent as in Fig. 1 and used for truck bolsters, and as we used about 24 of them a day, it was up to me to provide the means for an accurate, safe and economical method of doing the work.

Everything was going at top speed and as the wheel press was idle a good part of the time, I turned my attention to it to



TOOT'S TOOLS FOR WHEEL PRESS.

help me out of a very bad hole, and I must give it credit for the excellent results obtained from its slow and steady power and my happy thought for a form to do the bending.

I got two castings made, one to fit over the ram head and one for the channel, shaped just right, but slightly deeper than the bend at *A*. I took this precaution, for a blow with a sledge would take some of the bend out of the bar if too deep (and we always bent them a little more than needed). If not bent enough they had to be pressed up again. Fig. 2 shows this casting fitted on a scrap axle, which in operation swung in the hooks on the top bar.

The ram casting is shown in Fig. 3 with a heavy steel knife edge keyed in it and the working face rounded nicely so it would not cut the bar. A truck just the right height to keep the bars even, and a gauge to test the bends completed the outfit and—— Well, my troubles were a thing of the past on that job when I got this rig going.

There is plenty of such bending the wheel press can be made to do, if called on, for its slow powerful action is just suited for this kind of work, and the best feature is: you can run it up a foot or a sixteenth of an inch and stop it instantly, which is a feature that no other machine possesses. T. Toot.

St. Louis, Mo.

The Walschaerts Valve Gear.

Editor:

The Walschaerts Valve Gearing is becoming more popular every day, as is witnessed by the large number of locomotives now equipped with the gearing, and to those that are not familiar with this kind of valve motion, it perhaps would be well to know that, unlike the Stephenson shifting link, which has a vertical movement up and down upon the link block, the link with the Walschaerts gear is pivoted at one fixed central point and oscillates freely about that point, but has no vertical movement; and also that the radius rod carries the link block which slides up and down in the link slot.

The radius of the link is from point of connection between radius bar and combination lever, to center of link slot, and the curve of the link is that swept out by this radius.

With the Walschaerts gear, the lead is constant with reverse lever in any position; in other words, hooking up the lever will affect the cut-off, the same as with the Stephenson gear, but will not affect the lead, that always remaining constant.

With the Stephenson gear, lead is obtained by advancing eccentric the amount of the lap and lead ahead of a vertical line drawn at right angles to a horizontal line through center of axle, this advance being called the angular advance, while with the Walschaerts gear, center of eccentric crank pin is placed upon a vertical line drawn at right angles to center line of motion and cannot be advanced from that position. Therefore, there is no angular advance to obtain lead with this gear, but lead is obtained by, and the amount depends upon, leverage in combination lever and crosshead connection.

When it is desired to change the lead or set the valve, the first important move to make is to ascertain the correct length of eccentric rod, which should be of such length that the reverse lever can be moved from full gear forward to full back gear without moving the valve.

Having placed engine upon exact dead center, forward center preferable, and having all parts of gear connected, watch valve while assistant moves reverse lever as before mentioned. If eccentric rod is too long the valve will move back towards link when reverse lever is moved from forward gear to back gear, while if too short, the valve will move away from link.

Having found rod of proper length, or adjusted it to proper length, we proceed to set valve by disconnecting crosshead connection, and, with engine on exact dead center, pull or place the combination

ner, and Brown was wounded very seriously in the back.

"We got a wagon from the farmer, and had a tedious journey over the 35 or 36 miles to Vinita, and that night we went down to Ft. Smith and left our prisoners in jail there, and the following March they were tried before Judge Parker and he promptly found them guilty and sentenced them to the penitentiary for twenty-five years each. For some reason, unexplainable to us, Judge Parker paroled Bob Rodgers, the leader of the gang, but he went right out again, and was worse than he ever was before; he finally was killed, after he had killed three or four Deputy U. S. Marshals."

Driving Boxes, Shoes, Wedges and Rods

By a General Foreman.

To go into details covering locomotive driving boxes, shoes, wedges and rods, would make it necessary to take up each subject separately at considerable length. Each subject, however, is so closely identified with the others that in a general discussion it is well to bring them under one head. The most important requisite for the proper working of the machine is the spacing of the driving wheels, or the locating of the driving axles with their centers the exact distance apart on each side of the engine and the axles square with the frames.

While the operations which bring about these results are quite simple and can be readily understood by a workman of ordinary intelligence, we often find machinists who have not taken the time to consider the details sufficiently to understand the laying out of the frames, shoes and wedges. We know of shops where permanent centers were located on the main jaws of the frames so that when the engine received repairs again, these centers could be used in order to avoid the small amount of trouble necessary to lay them out again. Years ago, a locomotive frame was not so apt to get out of alignment to the same extent that it is to-day, and, in making repairs to it, we were more particular to know that all surfaces were machined. Now-a-days, we weld frames without removing from position, and, even though modern locomotives have heavy steel frames, we find distortion due to strains of operation.

To properly locate the centers of main axles, we should take a point as close to the back side of the cylinders as possible, find the center of frames and tram from that point to the center of the main jaws, making the marks on frame of equal distance from the center located at or near the cylinders. This should insure the main driving axle being at right angles to the longitudinal center of the locomotive, or as we generally term it "square" with the engine. From these centers, the centers are laid out on the other jaws and then each jaw is laid out to take care of the conditions affecting it. On some engines the laying out of main jaws must be on the inside of frames and then transferred to the outside surfaces. If this can be avoided, as we frequently find it can, the operation is simplified by making tram marks on the outside surface only. It goes without saying that in addition to having tram marks located proper distances longitudinally, that they should also be the same vertical distance from center of axle. Generally, this can be taken by measuring from tops of frame.

It was not so many years ago that it was the practice in some shops to lay out the shoes and wedges, put them in position and try each driving box in its jaw to make certain of correct workmanship. To-day, we cannot take so much time. We bore the driving box brasses so that the center is equally distant from the shoe and wedge face of box and the men who lay out shoes and wedges merely require the size of the box to go ahead with their work. If the boxes do not have brasses bored centrally, the operation is made more difficult as we must then locate the distance from the center of brass to shoe and wedge faces and use these measurements in laying out the shoes and wedges for planing instead of using one half the size of boxes as would be the case if brass were bored centrally. After shoes and wedges are planed, they should always be placed in position temporarily in order to tram them and know that the laying out and the planing have brought the desired effect.

Many railroads are using a brass shim on shoe and wedge faces of driving boxes in order to have a more easily lubricated surface and to avoid planing the boxes properly to parallel the two surfaces. There is a variation of opinion as to the benefits derived although the majority favors the brass shims on the steel boxes.

It is as important to have the side rods trammed to the proper length between centers as it is to have the driving axles spaced properly, the length between centers of axles of course being the same as the distance between crank pin centers of the same pairs of driving wheels. It takes but a slight variation between the rods and the shoes and wedges to make serious trouble, causing the engine to ride hard, making a jerk as it passes the dead centers, causing pins to run hot and contributing largely to broken axles, frames, crank pins, etc. Failure to have the engine in tram is responsible in the majority of cases for the wearing of flanges. Some engines will run on a crooked road with little flange wear, while others can be kept in service but a short time without renewal of tires or having them turned for proper flanges.

It is important for engineers to give these matters careful thought in order to more intelligently make out work reports, as they have the opportunity



QUEBEC FROM THE ANCIENT CITADEL.

ing in the application chamber the single pointer gauge will show these pressures.

As it was stated in the letter mentioned, the reduction should be almost continuous until the train has stopped on account of the ports m and n building up application chamber pressure; for, in this position of the brake valve, the signal system is no longer connected with the application chamber.

On account of the size of the port connecting the two chambers the pressure will be withdrawn from the application chamber faster than it can flow from the pressure chamber and if the pressure is allowed to increase in the application chamber it will move the main piston to its application position, closing the brake pipe exhaust and permitting more pressure to flow into the brake pipe.

The pressure in the equalizing reservoir of the automatic brake valve will leak past the equalizing piston packing ring to the atmosphere and in a short time the gauge will show no pressure in this reservoir, and as the independent brake valve handle is moved to make the application, a reduction in the application chamber will occur when the handle reaches running position, due to the pressure flowing to the equalizing reservoir. If this is found undesirable the automatic brake valve handle can be placed in service, or lap position.

This method of handling a train not only applies to a broken brake pipe, but also to a case where both the main reservoir and few valve pipes are broken; or where the automatic brake valve is disabled from a broken body or a twisted or broken rotary key, it will only be necessary to stop the main reservoir leak and proceed as previously stated.

This appears even simpler when the break occurs with the later H 6 equipment, as there are no ports m and n in the distributing valve, and in case of the broken brake pipe the handle of the automatic brake valve can be returned to service position which will prevent the flow of air from the feed valve pipe to the application chamber of the distributing valve.

When the brake valve is broken or all the pipes broken, so there is no air pressure on the brake valve the position of the handle is of no consequence.

When the brake valve cut-out cock is located in the main reservoir pipe with the H 6 equipment a check valve is placed in the pipe which conducts pressure from the rotary valve and feed valve pipe to the application chamber pipe during emergency applications.

This check valve prevents a backward flow of air which might unseat the rotary valve and allow the application chamber pressure to escape at times when there is no air pressure on top of the rotary valve which may occur when the engine is the second one in double-heading, should the brake valve on the first engine be allowed to remain in application position until all the brake pressure has escaped.

Therefore, it will be noted that if the H 6 brake equipment with the cut-out cock in the reservoir pipe is used to operate the train brake while there is no pressure in the automatic brake valve this check valve will prevent a flow of air from the application chamber.

When the stop-cock is located in the brake pipe of the H 6 equipment, this check valve is not used, and if the brake is to be operated under the conditions mentioned this branch pipe between the automatic brake valve and application chamber pipe should be blanked.

From the construction of the independent valve used with the H 6 brake valve, it will be noticed that when the handle is in running position the exhaust port of the distributing valve is open to the automatic brake valve and while the equalizing valve is in application position the exhaust port is lapped and no reduction in application chamber pressure will occur until the handle is moved to release position.

Referring again to the letter published in the October number it states that if the application chamber pipe is broken the application chamber leak could be plugged and the double heading pipe disconnected to form an exhaust for the application chamber pressure and the driver brake used during automatic applications with the H 5 brake equipment the application chamber pipe is the lower one while with the H 6 brake the application chamber pipe is the upper one and if broken and plugged the lower one should not be disturbed, as it would destroy the driver brake holding feature. To plug this pipe would be equivalent to plugging the exhaust port of a triple valve.

On account of the length of the brake pipe on modern locomotives some difficulty in obtaining full brake cylinder pressure on the train during emergency applications has been experienced when double-heading; for this reason the tenders on some locomotives are equipped with a quick action triple valve in addition to the E. T. brake.

The standard straight air brake double-seated check valve being used between the triple valve and brake cylinder with the pipe form the distributing valve connected to the straight air side.

When the tender is equipped in this manner it can be used with an engine equipped with the ordinary automatic brake with the combined automatic and straight air brake or with the E. T. brake.

When used with the E. T. brake the stop cock in the pipe leading from the distributing valve to the tender brake should be left open while the triple valve is in service.

When the triple valve is cut out the stop cock should also be closed, and if this cock is left closed when the triple valve is again cut in a slight leak past the gasket on the straight air side of the check valve will allow pressure to build up in the pipe between the check valve and stop cock during applications of the brake, and as the triple valve starts to reduce the pressure in the brake cylinder when the brake is released this stored up pressure will force the check valve over, cutting off the flow of air from the triple valve.

When the brake is stuck in this way it cannot be bled off; the only way it can be released is by opening the stop cock or disconnecting the hose between the engine and tender.

G. W. Kiehm.

Washington, D. C.

A Freak That Was Missed.

Editor:

I have been reading Sinclair's "History of the Development of the Locomotive Engine," and have been particularly interested in the chapter on Freak Locomotives; but I find that one celebrated freak has been omitted. That freak was known as the Swinerton Locomotive, and was very much in evidence in New England in 1888. The designer of that locomotive conceived the idea that driving wheels covered with flat facets would adhere more tenaciously to the rail than those with a true round surface. A locomotive was built to put that idea into practice, a single pair of 5-ft. driving wheels being used with 105 flat milled spots on the outside of the tire.

In looking over my bound volumes of The Locomotive Engineer I found that the editor made fun of the claims made for the Swinerton engine, and that for months he permitted the inventors to do their best with the locomotive without embarrassing them with criticism. Then he gave them a final blast as follows:

"When the Swinerton locomotive came tap-tapping along with 105 flat spots and 41,000 pounds on a single pair of drivers, this paper was taken to task by some of the Boston stockholders for condemning 'a new departure in locomotive practice before it could be tried.' So we agreed to keep still for six months, which we have done.

"We have watched this engine's performance, and find that she has spent as much time on the jacks as on the road; that her flat spots have got their corners knocked off, and that she has never made twenty-five miles at a stretch without getting her driving boxes too hot for safety. She has had some of the weight transferred from the drivers to other wheels, and new axle and driving boxes.

"The engine was built in defiance of

Air Brake Department

Broken Air Pipes.

In an article on this subject recently published in these columns it was stated that the subject would be continued, and it will be the object of this article to explain how the brakes on a train of cars



MISPLACED SWITCH DID IT.

can be operated from a locomotive equipped with the New York Air Brake Company's B² H. S. equipment when the brake valve branch of the brake pipe is broken between the brake pipe and the connection to the high-speed controller valve and how the brake can be operated should the brake valve itself be broken.

If the brake valve branch is broken off from the brake pipe when out on the road the handle can be placed on lap position and both ends of the break plugged, care being taken not to obstruct the flow of air through the brake pipe, the stop cocks to the brake cylinders can then be closed and if there are stop cocks to the auxiliary and truck brake reservoir they should also be closed.

The triple valve can then be taken apart and the triple valve piston removed, the back cap and gasket replaced in their proper position and the triple exhaust port plugged, which will establish a communication between the brake cylinder pipe and the brake pipe and prevent the escape of air at the triple exhaust.

The piston should then be removed from the double check valve and the pipe again connected up, and the adjusting nuts on the straight air reducing valve and the highspeed controller should be screwed down to create a greater tension on the springs than the brake pipe pressure to be used.

If these parts can be reached when the locomotive is out on the road this work can be done in a very few minutes' times and it may take several hours to secure another engine to haul the train.

If it can be done at all it can be done as quickly as most any other breakdown on a locomotive can be handled out on the road and when done the brake valve handle can be placed in release position and air will flow from the brake valve through the straight air reducing valve, double check valve and the triple valve into the brake pipe on the engine and train. The adjusting nut of the reducing valve is screwed down to prevent it from closing when 40 lbs. pressure is attained in the brake pipe, the duplex controller will govern this pressure.

The safety valve adjusting nut on the high-speed controller is screwed down to prevent it from exhausting brake pipe pressure when it reaches the figure for which the controller safety valve is adjusted. When it is desired to apply the brake the valve handle can be placed in running position and brake pipe pressure will flow from the brake pipe through the triple valve, brake cylinder pipe, double check valve, straight air reducing valve and the brake valve to the atmosphere applying the train brake. The flow of air through the triple valve would not be free enough to apply the brake in quick action, but when the shortest possible stop is desired, opening the lever on the high-speed controller would have a tendency to assist in reducing brake pipe pressure. The stop cocks to the reservoirs are closed so that their volumes are not added to the brake pipe volume.

BROKEN BRAKE PIPE IF THE TENDER IS EQUIPPED WITH THE COMBINED AUTOMATIC AND STRAIGHT AIR.

If the tender is equipped in this manner and the standard ¾ of an inch hose are used to connect the tender and driving brake cylinders, sometimes the standard brake pipe hose are used by bushing the angle fitting to ¾ of an inch, in either case if the brake cylinder pipe at the rear of the engine can be connected with the brake pipe on the front of the tender, this connection can be made in case of any broken brake pipe on the engine.

If the brake pipe is broken anywhere below the high-speed controller connection or anywhere above it, it will only be necessary to plug the break toward the brake valve, the rest of the brake pipe can be removed from the engine. but if the controller is also broken off the break must be plugged.

It will then only be necessary to cut out the driver brake triple valve, close the stop cocks to the driving brake cylinder, leaving the cock in the pipe to the tender cylinder open and when the connections between the engine and tender are made, placing the brake valve handle in release position will charge the train brake pipe through the engine brake cylinder pipes, and placing the handle in running position will reduce the pressure in these pipes and apply the brakes.

As in other cases the adjusting nuts on the reducing and controller valves must be screwed down to fully charge the brake pipe and prevent the escape of

General Foremen's Association.

(*Continued from page 296.*)

right side there must be some resistance or strain of some kind.

I have prepared sketches showing the right and left main crank pin in different position. Sketch No. 1 shows the right crank pin approaching the forward center and the left pin the top quarter. If the engine is working in the forward motion the power developed by the steam in the left cylinder is in the direction of the arrows shown on sketch. This would practically make the left front pedestal jaw a fulcrum between the left crank pin bearing and the right pedestal jaw. After the point of cut-off on the right side has been passed and the pressure in the back end of the right cylinder greatly reduced, whatever lost motion there might be in the right driving box or in the shoe or wedge, would be found at the points marked X in the sketch. Consequently when steam is admitted to the front end of the right cylinder no pounding would be heard, although there might be considerable lost motion in the right driving box. As the wheels revolve and the right pin approaches the bottom quarter, and the left pin the forward center as shown in sketch No. 2, a slightly different combination is found. The lost motion in the driving box will be at X, shown in sketch, and when steam enters front end of left cylinder there will be a pound if there is any lost motion in the left driving box. According to this view, it will be seen that as long as the engine is working in the forward motion the left side acts more or less as a slack adjuster for the right side.

This would give the right side an advantage over the left side. I believe this has more or less to do with, if it is not the direct cause, of the left side pound.

To illustrate what we found on one of our high-speed engines after having made 154,570 miles. This engine was received from the works December 28, 1905, and taken in shop for general repairs November 14, 1907. We found the left main shoe worn one-fourth inch and the right main shoe one-eighth inch. This is a very good illustration that there is something on the left side which causes this.

On November 20, 1907, we turned a Pacific Type engine 6416 out of back shop after receiving a general overhauling, and up to February 29, 1908, this engine made 26,826 miles. When this engine left back shop I personally spoke to the engineer in charge and told him to watch the left main driving box, which he did. The engineer in charge of this engine watched the driving boxes on this engine himself; he set up the left main wedge during the time I just mentioned one three-eighths inches, while he only set up the right main about five-eighths inches. I am positive that there was no pound on the left main driving box on this engine from the time she left back shop until February 29, 1908, but I attribute this to the carefulness of the engineer in watching the wedges of the left main driving box. However, I know that if the engineer in charge of this engine had not watched the left main driving box there would have been a pound.

I believe the illustrations I have given you will prove that there is something on the left side of an engine which causes this pounding. I am satisfied it is not in the counter-balancing of the wheel, as we have weighed the counter-balance and reciprocating parts on a Pacific Type engine and found them to be exactly correct.

Now, I have tried to give you all I know about the pounding of the left main driving box and I will be pleased to hear from all of you on this topic, as I think it is a very important one and should have technical consideration.

C. H. Voges,
Bellefontaine, O. N. Y. C. Lines.

This is a subject that has been before the railroad men for a good many years, and has been discussed, pro and con, by the railroad men all over the country.

A great many reasons have been assigned for it, but in my mind there is very little to be said on the subject. That it does exist, there is no question, and on all locomotives.

The right side of the locomotive being the lead, the lost motion is concentrated in the left, thereby producing a pound; the more lost motion, the greater the pound.

When the engine is put in the reverse motion the pound will transfer to the right side.

If the engine is properly taken care of, the lost motion kept up, such as wedges, driving box brasses and rod brasses, the pound is not sufficient to cause any great alarm, or do any great amount of damage.

It has been said by some that it was due to the engineer not taking as good care of the left side of the engine as he does of the right, but I do not think very much of this theory.

Then again, I think the counter-balancing of our engines has a great deal to do with the pounding. I think our present method of counter-balancing is to a great extent guesswork. For instance, an engine is counter-balanced so she runs very smoothly at a certain speed, but when you increase or decrease that speed, she does not run so smoothly and then the pound is more perceptible. I think if our engines were properly counter-balanced, the pound would be alleviated to a great extent.

E. R. Berry,
Galesburg, Ill. C., B. & Q. R. R.

There are several causes that we find which would result in making an engine pound in any one particular box. For instance if the crown brass was bored too large for the journal and put up in that condition, especially on the main pair of wheels, the engine would develop a pound almost immediately, although this would not show up so bad if on any other pair. The main boxes on any engine will begin to pound before any other box; that is if the same care has been taken in fitting brasses, shoes and wedges, etc. The reason for this is that the power is transmitted directly to this pair of wheels from the cylinder; they in turn transmitting the power to all the other drivers, thus putting a heavier strain on these boxes. The proper laying off of shoes and wedges has much to do with driving boxes pounding and so slightly it will cause a bad pound almost as soon as an engine is put in service. Especially is this true if the journal is flat through center line of motion or when the pin is a little below front center line. Also an engine that for any reason carries more steam on the left side than on the right will cause a pound in the left main box on account of the greater amount of friction caused by this side doing more work. I have often thought that excessive lead on one side of an engine, this lead of course being caused by a slipped eccentric or by putting up an eccentric in round house without running engine over to determine where it was set, will cause a pound on that side, in preference to the other. This is due to the greater pressure exerted on the pin at center line of motion as would be the case with the larger port opening at this point caused by excessive lead, causing journal to pound back and front of brass. I think that if more pains were taken in laying off shoes and wedges, getting engine and rods in perfect trim, also fitting driving boxes so there would be a bearing for nearly half a brass, and seeing to it that wedges were set up properly at all times, the pounding of driving boxes would be reduced to a minimum.

W. D. Kidneigh,
General Foreman, Union Pacific Ry.
Grand Island, Neb.

Modern Shop Construction.

Topic No. 2 was on modern shop construction. It deals with cross of longitudinal pits, location of wash rooms and lavatories, best location for each department, care of shop order material and convenience of storage. The subject was taken up by Messrs. L. H. Bryan, D. E. Barton and E. F. Fay. The report will be printed in a future issue. Topic 3 reporting work vs. engine inspection. Topic 4, the apprentice question will be taken later.